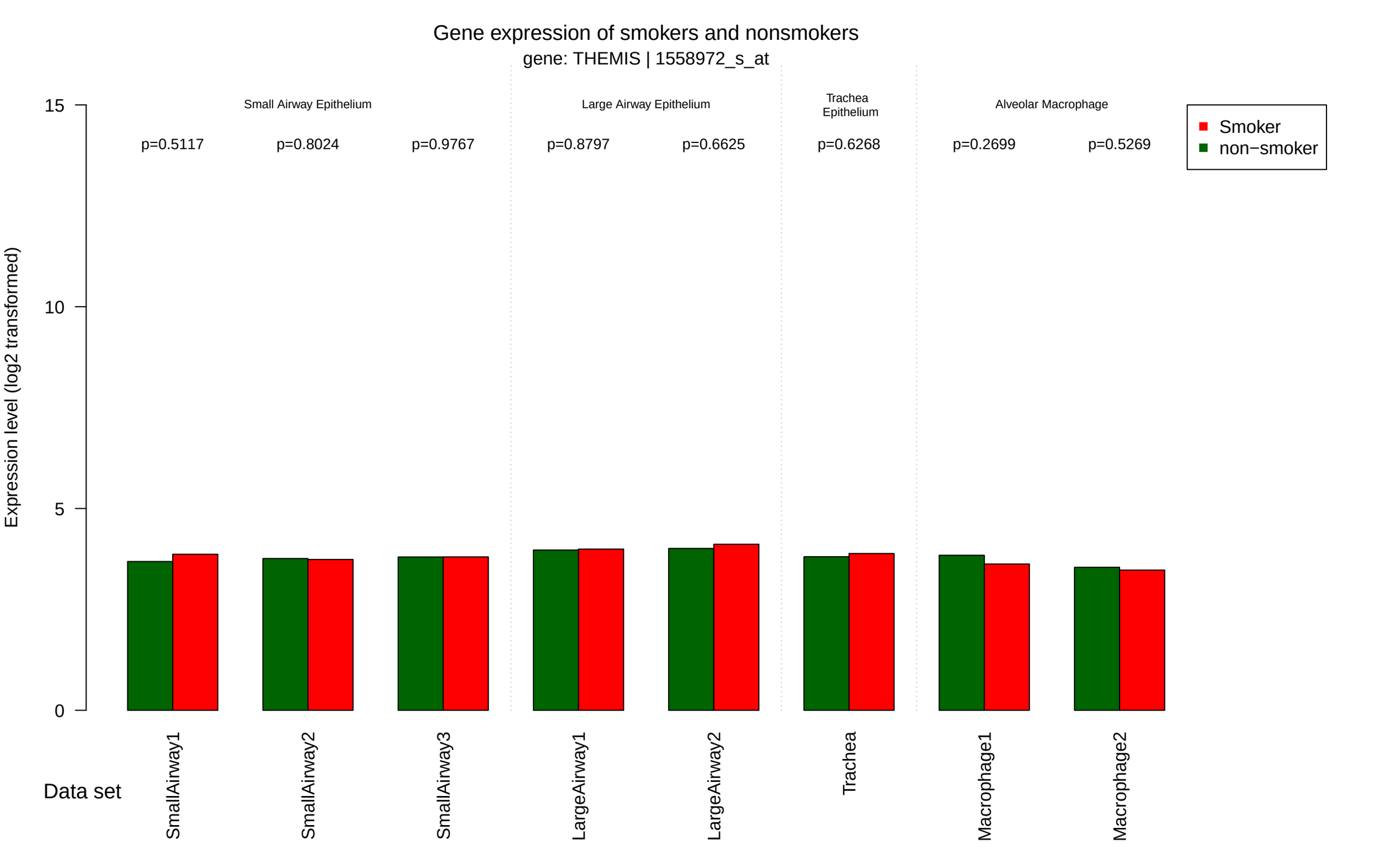
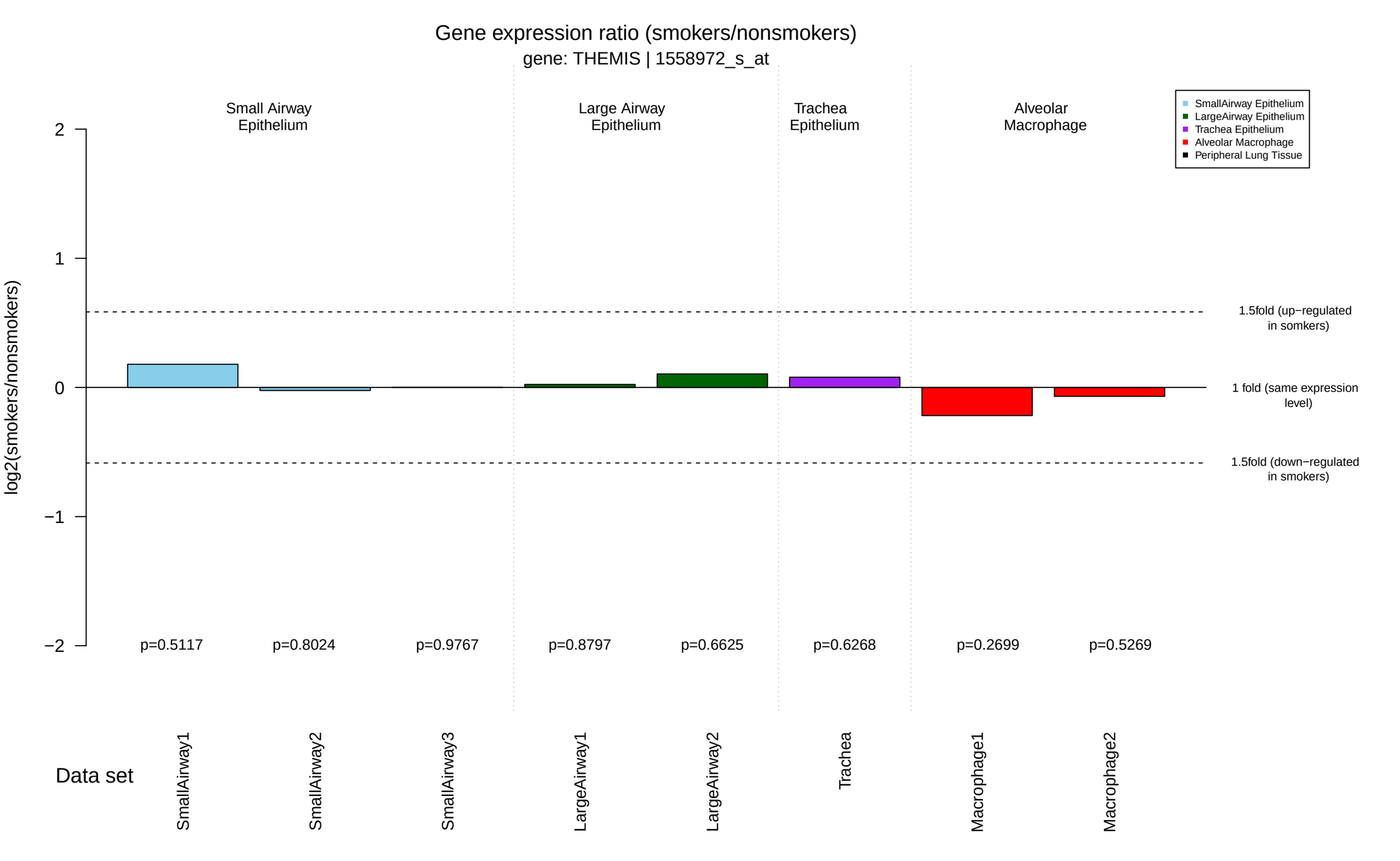
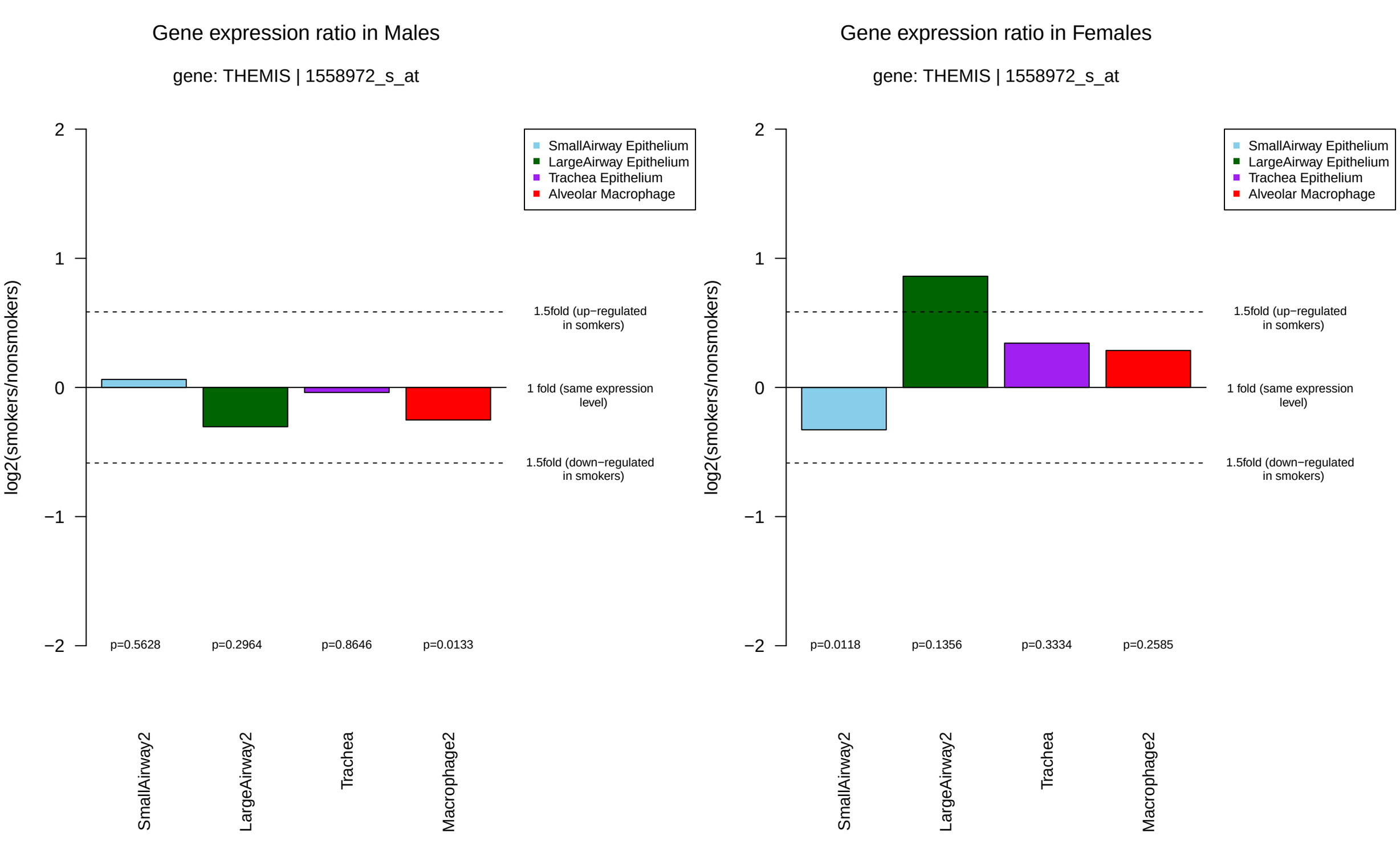
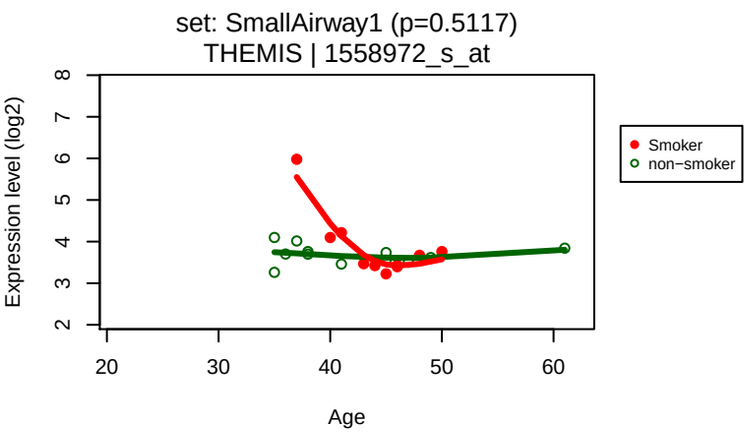
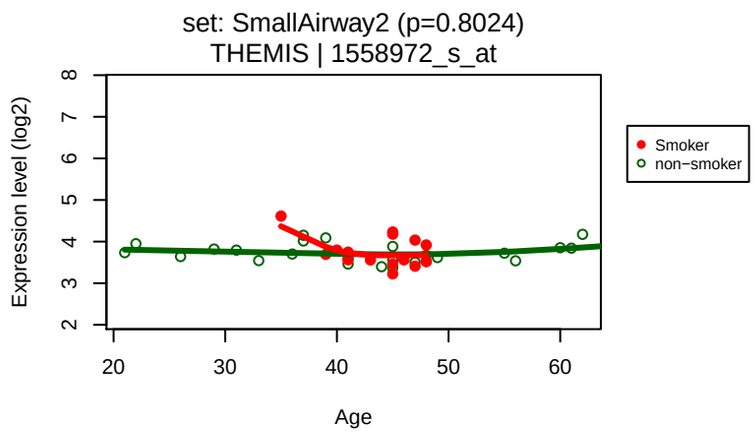
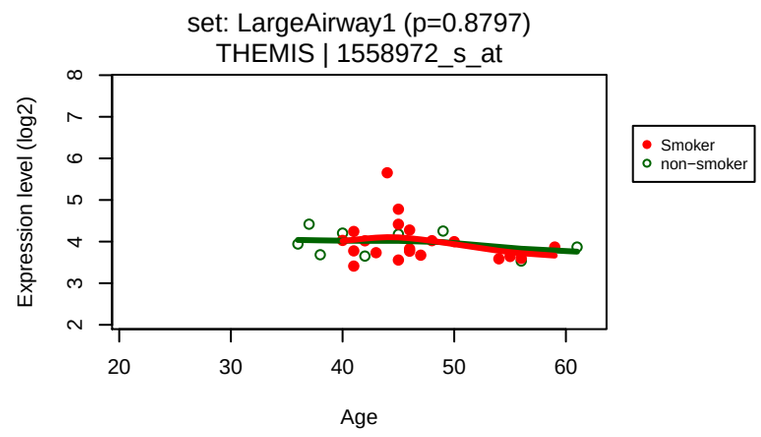

Gene expression of smokers and nonsmokers
gene: THEMIS | 1558972_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5117
p=0.8024
p=0.9767
p=0.8797
p=0.6625
p=0.6268
p=0.2699
p=0.5269
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

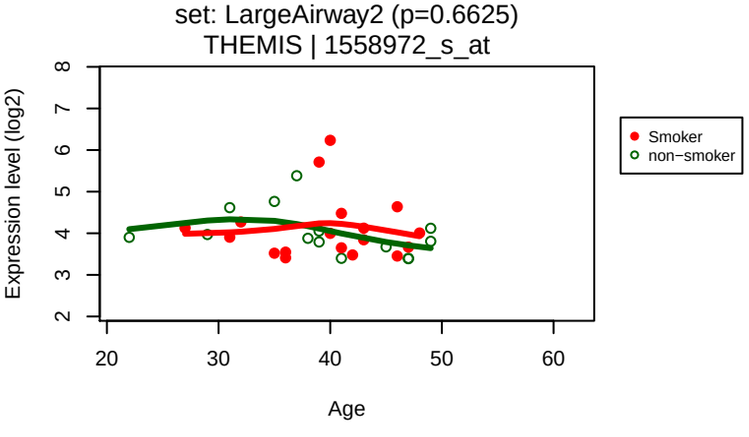

Gene expression ratio (smokers/nonsmokers)
gene: THEMIS | 1558972_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5117
p=0.8024
p=0.9767
p=0.8797
p=0.6625
p=0.6268
p=0.2699
p=0.5269
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

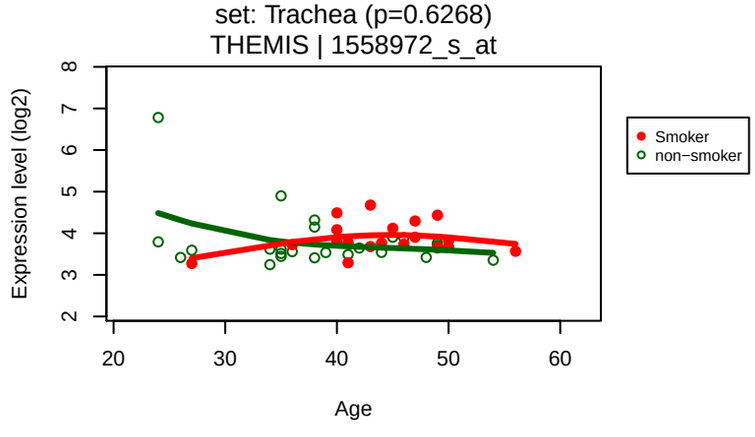

Gene expression ratio in Males
gene: THEMIS | 1558972_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5628
p=0.2964
p=0.8646
p=0.0133
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: THEMIS | 1558972_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0118
p=0.1356
p=0.3334
p=0.2585
SmallAirway2
LargeAirway2
Trachea
Macrophage2

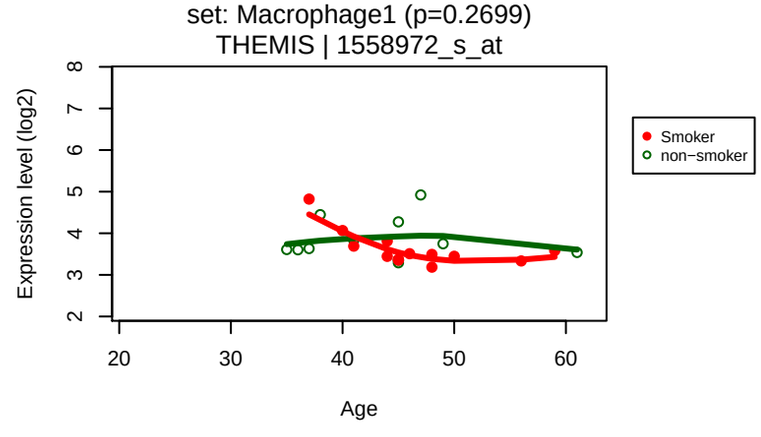

set: SmallAirway1 (p=0.5117)
THEMIS | 1558972_s_at
Expression level (log2)
Age
Smoker
non−smoker

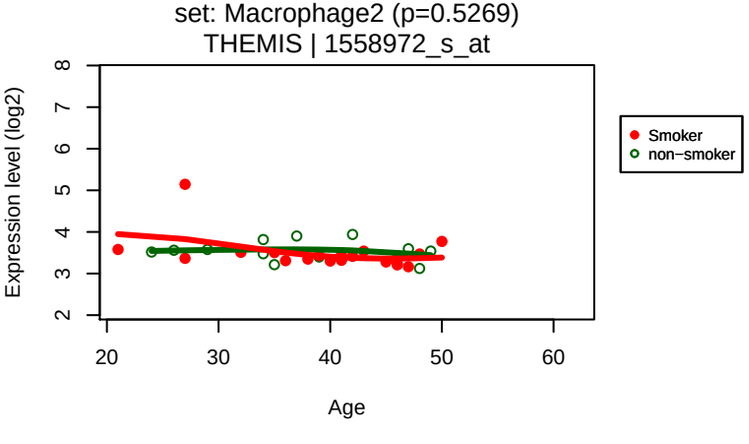

set: SmallAirway2 (p=0.8024)
THEMIS | 1558972_s_at
Expression level (log2)
Age
Smoker
non−smoker

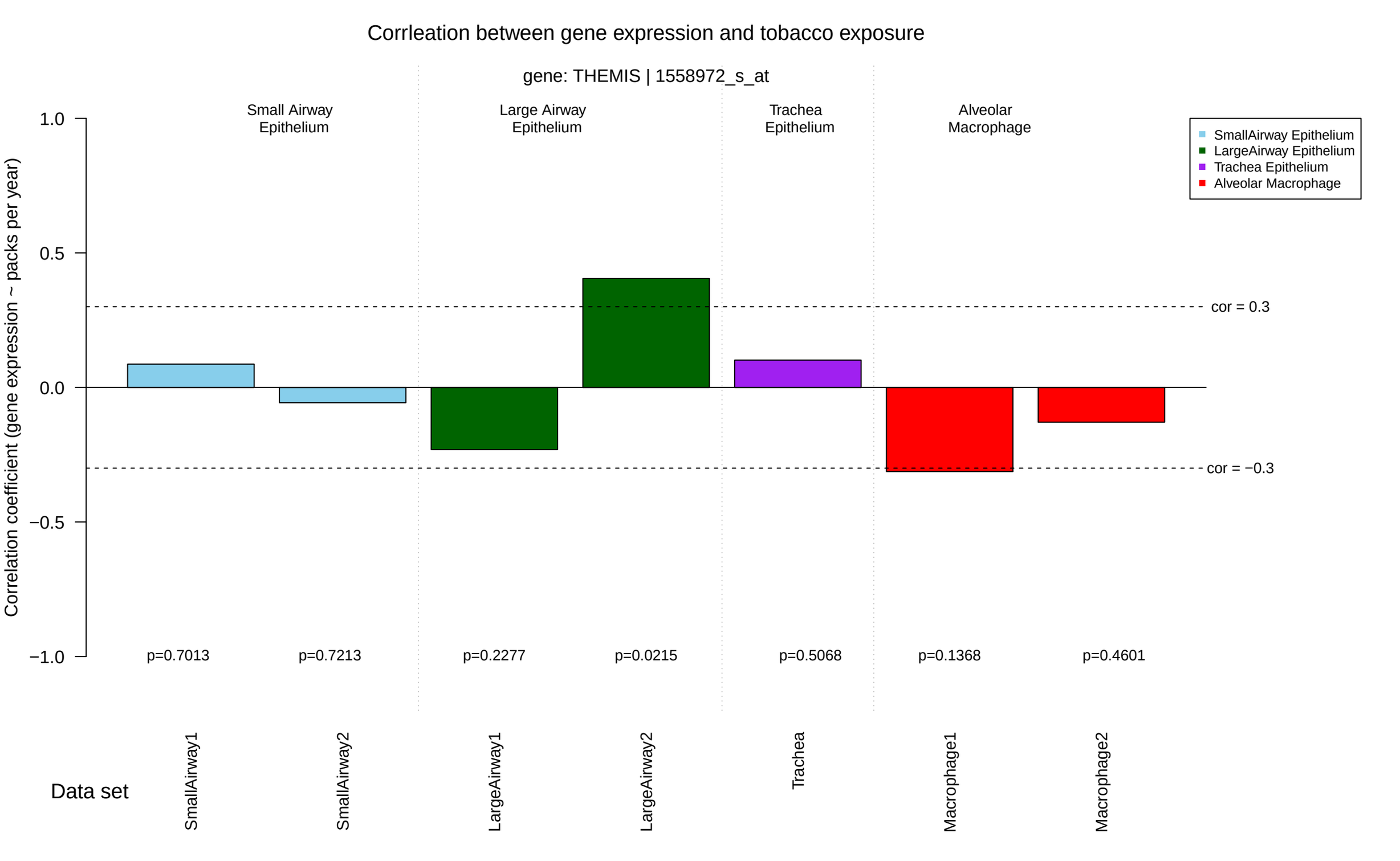

set: LargeAirway1 (p=0.8797)
THEMIS | 1558972_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6625)
THEMIS | 1558972_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6268)
THEMIS | 1558972_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2699)
THEMIS | 1558972_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5269)
THEMIS | 1558972_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: THEMIS | 1558972_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7013
p=0.7213
p=0.2277
p=0.0215
p=0.5068
p=0.1368
p=0.4601
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2